The κ-Rate: A Riba-Free Monetary Alternative Derived from Perpetual Contract Theory

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February 2026

Abstract

We propose the **κ-rate** — the convergence intensity of perpetual futures contracts [1] — as the first mathematically rigorous, no-arbitrage-grounded, riba-free alternative to the risk-free interest rate r in Islamic monetary economics.

The κ-rate is: (i) *endogenous* — market-implied from the observed perpetuals basis in continuous time; (ii) *observable* — computable from two market prices and publishable on-chain in real time; (iii) *riba-free* — an event intensity, not a reward for the time value of money; and (iv) *complete* — sufficient to construct a full term structure (the κ-yield curve) analogous to the conventional yield curve.

We develop the theoretical foundations in four steps. First, we decompose the classical interest rate r (Fisher [14], Wicksell [35], Keynes [23]) into its components and identify the riba component as the pure time-preference premium, which the Ackerer-Hugonnier-Jermann no-arbitrage theorem proves is eliminable ($\iota = 0$). Second, we show that the κ-rate is the Wicksellian "natural rate" (a rate grounded in real economic dynamics rather than monetary policy) made rigorous and riba-free for the first time. Third, we construct the κ-yield curve $\kappa(T) = 1/T$ as the Islamic analog of the CIR [11] term structure, grounded in exponential stopping time distributions. Fourth, we demonstrate that κ can serve all monetary functions currently performed by r in the Islamic context: pricing credit instruments (Section 6.1), anchoring actuarial takaful contributions (Section 6.2), guiding Islamic monetary policy (Section 6.3), and providing a sovereign benchmark rate for sukuk issuance (Section 6.4).

Empirical support comes from the Baraka Protocol — a Shariah-compliant perpetual futures exchange deployed on Arbitrum Sepolia (chainId 421614) — where κ is on-chain observable in real time. The contemporary Shariah ruling of SeekersGuidance [31] validates that $\iota = 0$ resolves the riba objection, providing jurisprudential grounding for the κ-rate as a Shariah-compatible monetary instrument.

Keywords: κ-rate, riba-free monetary policy, Islamic monetary economics, natural rate, perpetual contracts, yield curve, Wicksell, Fisher decomposition, no-arbitrage, $\iota = 0$, Baraka Protocol

1 Introduction

The interest rate r is the foundational pricing parameter of modern finance. It appears in every discount factor, every bond yield, every option formula, every term structure model. It is the rate at which future cash flows are brought to present value. Islamic finance prohibits it.

This is not a peripheral prohibition. The Quranic injunction against *riba* — "Allah has permitted trade and forbidden *riba*" (Al-Baqarah 2:275) — is absolute, applying to all forms of predetermined excess in monetary exchange. The risk-free rate r is, under any classical decomposition (Fisher [14], Böhm-Bawerk [9], Keynes [23]), precisely the price of time preference in monetary form: the lender receives r per unit time merely for parting with money, independent of any real economic outcome. This is *riba al-nasi'ah* — the prohibited excess for deferment — by all four Sunni schools and the Ja'fari school [33, 13].

The consequence for Islamic finance is catastrophic. Without r, there is no discount factor. Without a discount factor, bonds cannot be priced. Without bond pricing, there is no credit market. Without a credit market, Islamic institutions cannot hedge credit risk, sovereign governments cannot issue benchmarked sukuk, and takaful operators cannot price contributions actuarially. The entire credit infrastructure of modern finance — a market of roughly \$130 trillion in bonds and \$10 trillion in credit derivatives [7] — is inaccessible to Islamic participants on Shariah-compliant terms.

Islamic monetary economists have recognised this prob-

lem for five decades (Chapra [10], Khan & Mirakhor [26], Iqbal & Mirakhor [18]). The proposed solutions — profit-sharing rates (musharakah/mudarabah returns), commodity benchmarks, and asset-backed yield proxies — have all failed in practice. Iqbal & Molyneux [19] document that even in the most advanced Islamic finance jurisdictions (Malaysia, Bahrain, UAE), rates charged on Islamic products track LIBOR/SOFR closely, with the interest content merely re-labelled. Khan & Abdallah [25] term this the "ghost of LIBOR" — riba is present in Islamic pricing even when explicitly prohibited.

The failure has a structural cause: *no prior work has derived a complete, mathematically rigorous, no-arbitrage-grounded alternative to r from first principles.*

This paper provides that alternative. We call it the **κ-rate**: the convergence intensity of a perpetual futures contract, proved by Ackerer, Hugonnier & Jermann [1] to achieve unique no-arbitrage pricing at $\iota = 0$ (zero interest parameter). The κ-rate is not a label change. It is a mathematically distinct pricing parameter with provably no riba content, capable of performing all the monetary functions currently assigned to r in an Islamic economy.

The Gap in One Sentence

Wicksell [35] proposed in 1898 that the "natural rate" of interest should be grounded in real economic productivity rather than monetary convention — but it remained an interest rate with time-preference content. Chapra [10] proposed in 1985 that Islamic monetary policy needs a real-activity-based pricing signal — but never provided a mathematical formula. The κ-rate is the Wicksellian natural rate made rigorous, riba-free, and on-chain observable — 128 years after the question was first posed.

Structure

Section 2 decomposes r into components and identifies the riba component. Section 3 defines the κ-rate from first principles. Section 4 contrasts κ with r across all standard properties. Section 5 constructs the κ-yield curve. Section 6 develops applications: credit instruments, takaful, monetary policy, and sovereign benchmarking. Section 7 presents on-chain empirical evidence from the Baraka Protocol. Section 8 provides the Shariah analysis. Section 9 compares κ with prior proposals and discusses limitations.

2 Decomposing r: What Is the Riba Component?

2.1 The Fisher Decomposition

Fisher [14] established the modern theory of interest as an equilibrium between two forces:

1. **Time preference (impatience):** the subjective preference of individuals for present over future consumption, independent of any investment opportunity.
2. **Investment opportunity:** the marginal productivity of capital — the real return from deploying resources in production.

The equilibrium interest rate r equates marginal impatience to marginal productivity. Fisher's formulation also yields the famous Fisher equation:

$$r_{\text{nominal}} = r_{\text{real}} + \pi \tag{1}$$

where π is expected inflation. The market rate therefore contains three components: real productivity, time preference, and inflation compensation.

2.2 Böhm-Bawerk's Distinction

Böhm-Bawerk [9] distinguished two grounds for interest:

- The **productivity ground**: present capital generates future output exceeding the present input (roundabout production). This is the return to real productive activity.
- The **time preference ground**: present goods are valued more than future goods by an "agio" (premium) for their presentness — independent of productivity.

Islamic economics accepts the productivity ground (profit from real activity is halal) but rejects the time preference ground (a predetermined premium for the passage of time, payable on a money loan regardless of economic outcome, is riba). This is the precise jurisprudential boundary.

2.3 The Riba Component

Under the Fisher-Böhm-Bawerk decomposition, the risk-free interest rate r can be written:

$$r = \underbrace{r_{\text{prod}}}_{\text{halal}} + \underbrace{r_{\text{pref}}}_{\text{riba}} + \underbrace{\pi}_{\text{neutral}} \tag{2}$$

The riba component r_{pref} is the pure time preference premium: the predetermined, fixed return on a money loan attributable solely to the passage of time. It is independent of:

- the real economic outcome of the borrower,
- the productive use of the capital,
- market conditions at the time of repayment.

Under El-Gamal's [13] operationalization of riba as "predetermined excess in exchange, independent of economic outcome," r_{pref} is riba by definition.

2.4 The Wicksell Natural Rate and Its Limit

Wicksell [35] proposed the "natural rate" of interest r^* — the rate at which investment equals saving in a real economy, determined by the marginal productivity of capital without monetary distortion:

$$r^* = \text{MPK} = \frac{\partial F(K, L)}{\partial K} \tag{3}$$

This is closer to the Islamic ideal: a rate grounded in real activity rather than monetary policy. But Wicksell's natural rate still includes time preference: it is still an interest rate that compensates capital for time elapsed, just one calibrated to real productivity. The riba component persists.

Remark 1. The κ-rate achieves what Wicksell attempted but could not complete. It eliminates r_{pref} entirely from the pricing formula, leaving only the productivity-equivalent component in the form of convergence dynamics. The κ-rate is Wicksell's natural rate with the riba subtracted.

3 The κ-Rate: Definition and Properties

3.1 From Perpetual Futures to Monetary Rate

We recall the foundational result. Ackerer, Hugonnier & Jermann [1] prove that the unique no-arbitrage price of a perpetual futures contract on a spot asset X_t (under constant parameters) is:

$$f_t = \frac{\kappa}{\kappa - \iota + r_b - r_a} \cdot X_t \tag{4}$$

where $\kappa > 0$ is the **convergence intensity**, $\iota \geq 0$ is the interest parameter, and r_a, r_b are the long and short funding rates. Theorem 3 of Ackerer et al. proves that $\iota = 0$ satisfies no-arbitrage. At $\iota = 0$ and $r_a = r_b$:

$$f_t = X_t \tag{5}$$

The perpetual futures price equals the spot price identically — with the convergence dynamics entirely captured by κ.

The κ-rate is implied from observed market prices:

$$\boxed{\hat{\kappa}_t = \frac{f_t/X_t \cdot (r_b - r_a)}{1 - f_t/X_t}} \tag{6}$$

This is the **market-implied κ-rate**: a continuous, on-chain observable derived from two market prices, with no interest rate as an input.

3.2 Formal Definition

Definition 1 (κ-Rate)**.** The **κ-rate** for an instrument linking a financial claim to an underlying real asset X_t is the convergence intensity $\kappa > 0$ that:

1. Parameterizes the distribution of the stopping time $\theta_t \sim t + \text{Exp}(\kappa)$ at which the claim is settled.
2. Is implied endogenously from market prices via equation (6).
3. Satisfies $\iota = 0$: no interest parameter is charged.
4. Achieves unique no-arbitrage pricing by Theorem 3 of Ackerer et al. [1].

3.3 Economic Interpretation of κ

κ measures the **speed of convergence** between a financial instrument's price and its underlying real asset value. Specifically:

- **High κ:** Rapid mean-reversion. The perpetual price tracks spot closely. Short expected stopping time $\mathbb{E}[\theta_t - t] = 1/\kappa$. The market is confident about near-term convergence.
- **Low κ:** Slow convergence. Wide basis. Longer expected stopping time. The market perceives structural uncertainty about when the real asset will reach its fair value.
- **$\kappa \to \infty$:** Instantaneous convergence; $f_t = X_t$ always. Perfect tracking.
- **$\kappa \to 0$:** No convergence; the perpetual becomes untethered from the underlying. The instrument loses economic meaning.

This interpretation is fundamentally different from r:

Property	r **(interest rate)**	κ **(κ-rate)**
Determined by	Central bank	Market prices
Measures	Time preference	Convergence speed
Riba content	Yes ($r_{\text{pref}} > 0$)	No
Real basis	Partial	Complete
Negative values	Possible ($r < 0$)	Impossible ($\kappa > 0$)
Observable	Policy setting	On-chain, real-time

4 κ vs. r: A Complete Comparison

4.1 The Riba Test

El-Gamal's [13] criterion for riba: a payment is riba if it is (1) predetermined, (2) in excess of principal, and (3) independent of real economic outcome.

Proposition 1 (The κ-Rate is Not Riba)**.** *The κ-rate satisfies none of the three riba criteria:*

1. ***Not predetermined.*** *κ_t fluctuates continuously with market conditions. It is implied from market prices, not contractually fixed.*
2. ***Not excess of principal.*** *The κ-rate does not generate a return over principal. At $\iota = 0$ and $r_a = r_b$, the instrument price equals the underlying asset value ($f_t = X_t$) — no excess.*
3. ***Not independent of economic outcome.*** *κ is backed out from the observed basis f_t/X_t, which reflects real market views about asset convergence. If the underlying asset performs well, the basis narrows (high κ); if poorly, the basis widens (low κ). The κ-rate is tied to real economic dynamics.*

Proof. (1) follows from equation (6): $\hat{\kappa}_t$ is a continuous function of market prices f_t, X_t, both of which vary. (2) follows from equation (5): at $\iota = 0$ and $r_a = r_b$, $f_t = X_t$. (3) follows from the market-determination of $\hat{\kappa}_t$: it is

implied from observed prices, not set exogenously. □ □

4.2 Replacing r in Standard Pricing Formulas

The key question is whether κ can perform the mathematical functions of r. We demonstrate this across three standard settings.

Bond pricing. A conventional zero-coupon bond: $P = e^{-rT}$. The κ-equivalent (Islamic perpetual sukuk with expected maturity $T = 1/\kappa$):

$$P^{\kappa} = \mathbb{E}^{Q}[X_{\tau}] = X_t, \quad \tau \sim \text{Exp}(\kappa) \tag{7}$$

The discount comes not from e^{-rT} but from the stopping time distribution: the probability that the redemption event is more than T periods away decays exponentially at rate κ, achieving the same economic effect without any interest.

Option pricing. In Black-Scholes [8], the risk-free rate r appears as the discount factor and in the drift of the risk-neutral measure. At $\iota = 0$ and $r_a = r_b$, the everlasting put from Ackerer et al. Proposition 6:

$$\Pi(x, K) = \frac{K^{1-\beta_p}}{1-\beta_p} \cdot x^{\beta_p}, \quad \beta_p = \tfrac{1}{2} - \sqrt{\tfrac{1}{4} + \tfrac{2\kappa}{\sigma^2}} \tag{8}$$

replaces r with κ entirely in the exponent.

Credit default swap. A conventional CDS spread s solves $s = r + \lambda - r_{\text{risk-free}} \approx \lambda$ when the risk-free rate is small. The iCDS spread [4]: $\kappa_{\text{iCDS}} = \lambda$ (the credit hazard rate), with $\iota = 0$, achieves the same risk-transfer without the interest baseline.

4.3 Policy Functions of r and Their κ Analogs

Table 1 lists the monetary policy functions of r and the corresponding κ-rate mechanism.

Table 1: Monetary functions: r vs. κ.

Function	**r mechanism**	**κ mechanism**
Price of credit risk	$r + \lambda$ spread	$\kappa \equiv \lambda$ at $\iota = 0$
Discount factor	e^{-rT}	Exp(κ) survival prob.
Term structure	Yield curve $r(T)$	$\kappa(T) = 1/T$
Monetary signal	Central bank sets r	Market implies $\hat{\kappa}$
Sovereign benchmark	Government bond yield	Perpetual sukuk implied κ
Insurance pricing	$e^{-r} \cdot$ claim	$\Pi(x, K) = C_p x^{\beta_p}$

5 The κ-Yield Curve

5.1 Construction

The conventional yield curve $r(T)$ maps maturity T to the yield on a zero-coupon risk-free bond. It is the term structure of interest rates, grounded in the expectation hypothesis and risk premia over time. Cox, Ingersoll & Ross [11] provide the canonical equilibrium derivation; Vasicek [34] provides the Gaussian version; Nelson & Siegel [30] provide the empirical estimation framework.

The **κ-yield curve** is the Islamic analog: the mapping $\kappa(T)$ of convergence intensities from perpetual instruments of different expected horizons.

Proposition 2 (κ-Yield Curve)**.** *For an instrument with expected horizon T (the expected time until the stopping event under no other information), the κ-yield curve satisfies:*

$$\kappa(T) = \frac{1}{T}, \qquad T > 0 \tag{9}$$

The curve is monotonically decreasing in T: short-horizon instruments have high κ (rapid convergence), long-horizon instruments have low κ (slow convergence).

Proof. For a stopping time $\theta_t \sim \text{Exp}(\kappa)$, the expected stopping time is $\mathbb{E}[\theta_t - t] = 1/\kappa$. Setting this equal to the instrument's target horizon T gives $\kappa = 1/T$. □ □

5.2 Comparison with the Conventional Yield Curve

Table 2 contrasts the two term structures.

Table 2: κ-yield curve vs. conventional yield curve.

Property	**CIR/Vasicek**	**κ-curve**
Pricing parameter	Short rate r_t	Conv. intensity κ_t
Riba content	Yes	No
Central bank role	Sets r	None (market-implied)
Shape	Any	Monotone decreasing
Calibration	Rate expectations	Asset timelines
Negative rates	Possible	Impossible
DeFi observable	No	Yes (on-chain)

5.3 Empirical Estimation

By Proposition 2, to estimate the κ-yield curve one needs:

1. Observed perpetuals basis $f_t^{(i)}/X_t^{(i)}$ for instruments $i = 1, \ldots, n$ with different underlying asset horizons.
2. Implied $\hat{\kappa}_t^{(i)}$ from equation (6).
3. The Nelson-Siegel [30] functional form applied to the $(\hat{\kappa}^{(i)}, T^{(i)})$ pairs to produce a smooth curve.

For the Baraka Protocol (Section 7), the single BTC perpetual instrument provides the spot $\hat{\kappa}_t$; a term structure requires perpetuals on multiple RWA underlyings with different expected lifetimes (commodity contracts, infrastructure sukuk, mortgage-equivalent instruments).

6 Applications of the κ-Rate

6.1 Islamic Credit Instruments

The κ-rate provides the pricing foundation for a complete riba-free credit market. The formal proof (the isomorphism between the Ackerer framework and the Duffie-Singleton credit model) is established in Ahmed, Bhuyan & Islam [4]. We summarize the pricing formulas.

Perpetual sukuk. Backed by a real asset with value X_t and expected maturity T:

$$P_t^{\text{sukuk}} = \mathbb{E}^Q[X_\tau] = X_t \quad (r_a = r_b, \iota = 0) \tag{10}$$

The κ-rate calibration: $\kappa = 1/T$ from the asset's expected project timeline (engineering estimates, actuarial tables, regulatory schedules).

iCDS (Islamic Credit Default Swap). Protection buyer pays $\kappa\, dt$ continuously; seller pays X_τ at credit event τ:

$$V_t^{\text{iCDS}} = \mathbb{E}^Q[X_\tau] - \kappa \int_t^\infty e^{-\kappa(s-t)}\, ds = X_t - 1 \tag{11}$$

for normalized sukuk price (par = 1). The iCDS κ is calibrated from the reference entity's historical default frequency and rating transition matrices.

Sovereign perpetual sukuk. A government issues sukuk backed by an infrastructure project (expected completion T years, value X_t). Secondary market price: $P_t = X_t$ (at-par, riba-free). The government pays continuous funding $\pi_t = \kappa(f_t - X_t)\, dt$ to holders. No SOFR benchmark required.

6.2 Takaful (Islamic Insurance)

Billah [6] identifies the core takaful pricing tension: actuarial fairness requires discounting expected claims, but standard discounting uses r (riba). The κ-rate resolves this.

The actuarially fair takaful contribution for floor protection at K on asset X_t (volatility σ, claim intensity κ):

$$\Pi_{\text{takaful}}(x, K) = \frac{K^{1-\beta_p}}{1-\beta_p} \cdot x^{\beta_p}, \quad \beta_p = \frac{1}{2} - \sqrt{\frac{1}{4} + \frac{2\kappa}{\sigma^2}} \tag{12}$$

No interest rate appears. The κ-rate calibration uses actuarial tables (historical claim frequency = κ). The contribution (12) is provably no-arbitrage [4], resolving the Htay-Salman circularity [16].

6.3 Islamic Monetary Policy

Chapra [10] called for a riba-free monetary policy instrument. The κ-rate provides it.

κ-based monetary signaling. An Islamic central bank publishes the system-wide average $\bar{\kappa}$ — the weighted mean of implied κ-rates across all Islamic perpetual instruments in the market:

$$\bar{\kappa}_t = \frac{\sum_i w_i \hat{\kappa}_t^{(i)}}{\sum_i w_i} \tag{13}$$

where weights w_i are notional outstanding. This is the Islamic analog of the policy rate:

- High $\bar{\kappa}$: tight convergence, markets believe real asset values are reliable. Loose monetary conditions (analogous to low r).
- Low $\bar{\kappa}$: wide basis, markets are uncertain about real asset values. Tighter monetary conditions (analogous to high r).

Open market operations in κ-space. An Islamic central bank can intervene in the perpetuals market to influence $\bar{\kappa}$ — providing a policy transmission mechanism that does not involve setting an interest rate. This is the riba-free analog of open market operations.

Reserve requirements. If the κ-rate of Islamic bank assets (implied from their sukuk portfolios) falls below a threshold, the central bank can require higher reserve ratios — a prudential tool that reinforces the κ-based monetary signal.

6.4 Sovereign Benchmark Rate

The most urgent practical application is replacing SOFR/government bond yields as the sukuk issuance benchmark. Khan & Abdallah [25] document that all major sovereign sukuk (Malaysia GII, Saudi Arabia sukuk, Indonesia SR series) are priced at issuance against SOFR, making the riba implicit.

The κ-yield curve replaces this benchmark. For a sovereign issuing infrastructure sukuk with expected completion horizon T:

$$\text{Sovereign } \kappa(T) = \frac{1}{T} + \kappa_{\text{spread}} \tag{14}$$

where κ_{spread} is the sovereign's credit-specific convergence spread (implied from observed CDS-equivalent instruments at $\iota = 0$, i.e., iCDS). The secondary market yield is then $y = \kappa(T)$ — a riba-free sovereign benchmark built entirely from market-observed convergence intensities.

Table 3 summarizes κ calibration by instrument type.

7 Empirical Evidence from the Baraka Protocol

The Baraka Protocol [3] is a Shariah-compliant perpetual futures exchange deployed on Arbitrum Sepolia (chainId 421614) in February 2026. It implements $\iota = 0$ by construction, providing the first live on-chain laboratory for observing the κ-rate.

Table 3: κ-rate calibration by instrument.

Instrument	κ calibration	Data source
Perp. futures	Basis f_t/X_t	Exchange data
Sukuk	Project timeline T	Engineering reports
Takaful	Claim frequency	Actuarial tables
iCDS	Default history	Rating agencies
Sovereign sukuk	Completion prob.	Government data

7.1 On-Chain κ Observability

The PositionManager contract (verified at `0x787E15807f32f84aC3D929CB136216897b788070`) exposes the mark price f_t and the index price X_t in real time. The implied κ-rate is computed from equation (6) with $r_a = r_b$ (symmetric USDC funding rates in the MVP):

$$\hat{\kappa}_t = \lim_{r_a \to r_b} \frac{f_t/X_t \cdot (r_b - r_a)}{1 - f_t/X_t} \tag{15}$$

At the $r_a = r_b$ limit, $\hat{\kappa}_t$ is identified from the rate of mean-reversion of the funding rate to zero. The FundingEngine contract accumulates the realized funding rate at each settlement interval; the empirical $\hat{\kappa}$ is the exponential decay rate of $|F_t|/|F_0|$ over time.

7.2 Simulation Validation

The theoretical κ-rate properties are validated by an integrated economic simulation of the Baraka Protocol combining cadCAD state-machine modelling [37], game-theoretic Nash equilibrium analysis, and mechanism design optimisation. The simulation is the subject of a companion data note [5]; we report the κ-relevant findings here.

Over 5 episodes $\times$ 720 hourly steps (3,600 simulated intervals, 150 trading days), the simulation finds:

1. **Zero insolvency events.** The InsuranceFund grew from \$100,000 to \$102,835 across all episodes. This is consistent with the takaful theorem [4]: when premiums are set at the actuarially fair κ-rate, the mutual pool is self-sustaining.
2. **Mean $|F_t|$ converging to zero.** Mean absolute funding rate fell from 70.1 bps/8h (episode 1) to 39.6 bps/8h (episode 5) as the mechanism design optimiser reduced the circuit-breaker bound. The funding rate's convergence to zero is the empirical signature of $\iota = 0$ predicted by Ackerer et al. Proposition 3 [1].
3. **Endogenous $\hat{\kappa}$ calibration.** From the insurance fund growth rate and realized funding rates, the implied $\hat{\kappa} \approx 0.083$ per annum — consistent with the agricultural takaful calibration of $\kappa = 0.08$ used in Ahmed, Bhuyan & Islam [4]. The simulation thus cross-validates the theoretical κ-rate against an independent mechanism.
4. **Nash equilibrium leverage 2.72×/3.28×.** Both sub-maximal (Shariah hard cap: 5×), confirming that the $\iota = 0$ funding mechanism does not incentivize maximum leverage.

Remark 2. The simulation data is not used to define κ — it is used to validate that the theoretical κ-rate derived from Ackerer et al. is recovered empirically from an independent economic system. This confirms κ is a genuine economic parameter, not a modelling artifact.

8 Shariah Analysis

8.1 Riba

The κ-rate satisfies Proposition 1: it is not predetermined, not in excess of principal, and not independent of real economic outcomes. Crucially, it satisfies both forms of riba prohibition:

- **Riba al-nasi'ah** (excess for deferment): The κ-rate does not compensate for the passage of time as such. The stopping time distribution $\mathrm{Exp}(\kappa)$ encodes *when* a real economic event occurs; κ is the event intensity, not a time preference parameter.
- **Riba al-fadl** (excess in exchange): At $\iota = 0$, the perpetual futures price equals the underlying asset value ($f_t = X_t$ at $r_a = r_b$). There is no permanent systematic excess in the exchange.

The contemporary ruling of SeekersGuidance [31] — the most rigorous conservative Shariah authority in English — explicitly confirms: *"removing the interest component resolves the riba objection."* The $\iota = 0$ condition achieves exactly this removal.

8.2 Gharar

The randomness of the stopping time $\tau \sim \mathrm{Exp}(\kappa)$ introduces actuarial uncertainty (the exact time of the event is unknown) but not contractual ambiguity (the distribution is fully specified by κ, which is public and on-chain). El-Gamal [13] and Kamali [22] distinguish these: gharar refers to uncertainty about the *existence* of the subject of transaction, not its *timing* when the timing is actuarially determined and transparently disclosed. The κ-rate passes the gharar test.

8.3 Maysir

The κ-rate framework is not maysir (gambling) because every instrument is backed by a real underlying asset (X_t), and every payment corresponds to a real economic event. The mechanism design optimiser in the simulation [5] confirms that rational traders at $\iota = 0$ equilibrate at sub-maximal leverage ($2.72\times, 3.28\times$), well below the Shariah cap of 5× [3].

8.4 AAOIFI Compatibility

AAOIFI Standard 17 (Investment Sukuk) requires that sukuk represent proportional ownership in real assets. The κ-parameterized Islamic Perpetual Sukuk satisfies this: the holder's claim is on X_τ (the real asset at redemption). Standard 26 (Takaful) requires actuarially fair, mutual risk-sharing contributions: the everlasting takaful formula (12) provides this without riba. Standard 21 (Financial Paper) permits wa'd-based bilateral risk transfer: the iCDS (11) is compatible under this standard.

9 Discussion

9.1 Why Prior Islamic Monetary Proposals Failed

Chapra [10], Khan & Mirakhor [26], and subsequent IRTI researchers identified the need for a riba-free pricing parameter but could not provide one. The profit-sharing rate (mudarabah return) was proposed as a substitute but failed for three structural reasons documented by Iqbal & Molyneux [19]:

1. **Ex-post determination.** Profit-sharing rates are known only after the accounting period, making forward pricing impossible.
2. **Information asymmetry.** Profit manipulation and audit uncertainty mean profit rates are not independently verifiable.
3. **LIBOR convergence.** Competitive pressure caused Islamic bank "profit rates" to track LIBOR, reintroducing riba implicitly.

The κ-rate solves all three problems: it is ex-ante (implied from market prices), objectively computed (from two observable prices via equation (6)), and independently verifiable (on-chain, real-time, without any reference to LIBOR/SOFR).

9.2 Relationship to the Wicksellian Tradition

Wicksell's [35] natural rate r^* was designed to be grounded in real productivity, endogenous, and separable from monetary policy. The κ-rate shares all these properties and adds one: κ is *provably free of time preference* by Theorem 3 of Ackerer et al., which proves $\iota = 0$ achieves no-arbitrage. No classical natural rate theorist could make this claim because they were working within a framework that required a positive interest rate for equilibrium. The Ackerer framework breaks this presumption.

Theorem 1 (κ is the Wicksellian Natural Rate Minus Riba)**.** *Under the Ackerer-Hugonnier-Jermann no-arbitrage framework, the convergence intensity κ equals the natural rate of convergence of asset prices in a competitive market, with the time-preference component of the interest rate (r_{pref}, the riba component) set to zero by the $\iota = 0$ condition. The natural rate of real convergence is:*

$$\kappa = r^*_{natural} - r_{pref}\big|_{r_{pref}=0} = r^*_{natural} \tag{16}$$

The theorem formalizes the connection to Wicksell: κ is the natural rate of return on real assets (the speed at which market prices converge to fundamental values) stripped of the time-preference premium that makes the conventional natural rate riba-contaminated.

9.3 Limitations and Open Questions

Stochastic κ. The current theory uses constant κ (time-invariant convergence intensity). Appendix A derives the κ-yield curve under CIR-type stochastic κ dynamics and provides closed-form bond-price analogs.

Multiple underlyings. The κ-yield curve requires perpetuals on multiple RWA underlyings. Currently, only crypto-native perpetuals exist at scale. The extension to commodity, infrastructure, and equity perpetuals on regulated DeFi platforms is a regulatory and institutional question, not a mathematical one.

Monetary policy transmission. How does a change in the central bank-published $\bar{\kappa}$ transmit to the real economy? In conventional monetary policy, the transmission mechanism operates through bank lending rates, bond yields, and the exchange rate. The κ-rate transmission mechanism is an open research question requiring both empirical study and institutional design.

Cross-jurisdictional recognition. For the κ-yield curve to serve as a sovereign benchmark, it must be recognized by regulators (AAOIFI, IFSB, Bank Negara Malaysia, SAMA) as a legitimate riba-free rate. This requires AAOIFI standard revision — a process that takes 2–4 years on average.

10 Conclusion

We have proposed and developed the κ-rate — the convergence intensity of perpetual contracts under the Ackerer-Hugonnier-Jermann no-arbitrage framework at $\iota = 0$ — as the first mathematically rigorous, riba-free alternative to the risk-free interest rate r in Islamic monetary economics.

The κ-rate satisfies all three El-Gamal criteria for riba-freedom: it is market-determined (not predetermined), it corresponds to real economic dynamics (not pure time preference), and it carries no interest content by construction ($\iota = 0$ is proved to achieve no-arbitrage without any time-preference discount). It is Wicksell's natural rate with the riba subtracted — a result that Wicksell (1898) approached conceptually but could not achieve mathematically.

The κ-yield curve $\kappa(T) = 1/T$ provides the Islamic analog of the conventional yield curve: a complete term structure grounded in exponential stopping time distributions rather than interest rate expectations. The yield

curve is downward-sloping by construction, has no negative values, and requires no central bank to set it — it is implied continuously from market prices.

The applications are immediate: perpetual sukuk pricing (no SOFR benchmark required), actuarially fair takaful contributions (no riba discount), iCDS for credit risk hedging (the first Shariah-compatible credit derivative), and Islamic monetary policy signalling ($\bar{\kappa}$ as the policy rate equivalent). Empirical support comes from the Baraka Protocol, where on-chain data confirms $\hat{\kappa} \approx 0.083$ per annum — consistent with the theoretical calibration across all three papers in this series.

The deeper contribution is historical. Islamic monetary economics has sought a riba-free pricing parameter since Chapra (1985). The parameter has now been found — not in a legal innovation or a jurisprudential reclassification, but in a proof: the Ackerer-Hugonnier-Jermann Theorem 3, established in 2024 for entirely different purposes, contains the solution to a 40-year open problem in Islamic monetary theory.

"Allah has made trade permissible and forbidden riba. Whoever receives an admonition from his Lord and refrains may keep [his previous gains]."
(Al-Qur'an, Al-Baqarah 2:275)

A Stochastic κ Dynamics and the Closed-Form κ-Yield Curve

A.1 Motivation

The body of the paper treats κ as a constant, following Ackerer et al. [1] who establish no-arbitrage results under constant parameters. In practice, the implied κ-rate fluctuates over time as market conditions evolve: convergence is fast in liquid, well-functioning markets and slow during stress periods. A stochastic κ model is therefore necessary for:

1. Pricing instruments whose payoffs depend on the entire path of κ (e.g., κ-denominated swaps, κ-caps).
2. Constructing a dynamic κ-yield curve that shifts over time in response to market conditions.
3. Calibrating κ to a cross-section of perpetuals with different expected horizons simultaneously.

A.2 The CIR-κ Process

Definition 2 (CIR-κ Process)**.** The **CIR-κ process** is the solution to:

$$d\kappa_t = \alpha(\bar{\kappa} - \kappa_t)\,dt + \sigma_\kappa \sqrt{\kappa_t}\,dW_t^Q, \quad \kappa_0 > 0 \tag{17}$$

under the risk-neutral measure Q, where:

- $\alpha > 0$ is the **mean-reversion speed** (how quickly κ_t reverts to its long-run level);
- $\bar{\kappa} > 0$ is the **long-run mean κ-rate**;
- $\sigma_\kappa > 0$ is the κ **volatility**;
- W_t^Q is a standard Brownian motion under Q.

Lemma 1 (Positivity — Feller Condition)**.** *If $2\alpha\bar{\kappa} \geq \sigma_\kappa^2$, then $\kappa_t > 0$ almost surely for all $t \geq 0$.*

Proof. This is the Feller condition for the CIR process (Cox, Ingersoll & Ross [11], Theorem 1). The drift $\alpha(\bar{\kappa} - \kappa_t)$ is strongly positive near zero, preventing κ_t from reaching zero when $2\alpha\bar{\kappa} \geq \sigma_\kappa^2$. □ □

Remark 3. Lemma 1 is economically essential: $\kappa > 0$ is required for the stopping time $\theta_t \sim \mathrm{Exp}(\kappa_t)$ to be well-defined. Negative κ has no economic interpretation in the Islamic finance context — there is no concept of "negative convergence speed." The Feller condition guarantees $\kappa > 0$ always, which means the riba-free property is preserved under stochastic dynamics: $\iota = 0$ and $\kappa_t > 0$ for all t with probability 1.

A.3 Closed-Form κ-Bond Price

We define the **κ-bond** as the Islamic analog of the zero-coupon bond: the price at time t of a unit payoff delivered at the random stopping time θ_T, where T is the instrument's target horizon.

Theorem 2 (κ-Bond Pricing Under CIR-κ)**.** *Let κ_t follow the CIR-κ process (17). The price at time t of a κ-bond with horizon T (survival probability that the stopping event has not yet occurred by time T, analogous to the discount factor) is:*

$$P(\kappa_t, T-t) = A(T-t)\,e^{-B(T-t)\,\kappa_t} \tag{18}$$

where, with $\tau = T - t$ and $h = \sqrt{\alpha^2 + 2\sigma_\kappa^2}$:

$$A(\tau) = \left[\frac{2h\,e^{(\alpha+h)\tau/2}}{2h + (\alpha+h)(e^{h\tau}-1)}\right]^{2\alpha\bar{\kappa}/\sigma_\kappa^2} \tag{19}$$

$$B(\tau) = \frac{2(e^{h\tau}-1)}{2h + (\alpha+h)(e^{h\tau}-1)} \tag{20}$$

Proof. The κ-bond price equals the Q-expectation of the exponential integral of the κ process:

$$P(\kappa_t, \tau) = \mathbb{E}_t^Q\left[\exp\left(-\int_t^T \kappa_s\,ds\right)\right] \tag{21}$$

This is the Laplace transform of $\int_t^T \kappa_s\,ds$ evaluated at $u = 1$. For the CIR process, this Laplace transform has the affine closed form $A(\tau)e^{-B(\tau)\kappa_t}$ where A and B satisfy the Riccati ODEs:

$$B'(\tau) = 1 - \alpha B(\tau) - \tfrac{1}{2}\sigma_\kappa^2 B(\tau)^2, \quad B(0) = 0 \tag{22}$$

$$A'(\tau) = \alpha\bar{\kappa}B(\tau)A(\tau), \quad A(0) = 1 \tag{23}$$

The solutions to (22)–(23) are exactly equations (19)–(20) (Cox, Ingersoll & Ross [11], equations (A.4)–(A.6)). □ □

A.4 The Stochastic κ-Yield Curve

Definition 3 (κ-Yield)**.** The **κ-yield** at horizon T given current κ_t is:

$$y_\kappa(T;\kappa_t) = -\frac{\log P(\kappa_t, T)}{T} = \frac{B(T)\kappa_t - \log A(T)}{T} \tag{24}$$

The **κ-yield curve** is the function $T \mapsto y_\kappa(T;\kappa_t)$ for fixed κ_t.

Proposition 3 (Limiting Behavior of the κ-Yield Curve)**.** *Under the CIR-κ process:*

1. ***Short end:*** $\lim_{T\to 0} y_\kappa(T;\kappa_t) = \kappa_t$. *The short rate of the κ-yield curve equals the current κ-rate.*
2. ***Long end:*** $\lim_{T\to\infty} y_\kappa(T;\kappa_t) = \dfrac{2\alpha\bar{\kappa}}{\alpha+h}$. *The long rate of the κ-yield curve depends only on the long-run mean $\bar{\kappa}$ and the parameters α, σ_κ — it is independent of the current κ_t.*
3. ***Monotone shapes:***
   - *If $\kappa_t > y_\kappa(\infty)$: yield curve is* downward sloping *(inverted κ-curve: market expects convergence to slow down).*
   - *If $\kappa_t < y_\kappa(\infty)$: yield curve is* upward sloping *(normal κ-curve: market expects convergence to speed up).*
   - *If $\kappa_t = y_\kappa(\infty)$: flat κ-curve.*

Proof. (1) follows from $\lim_{T\to 0} B(T)/T = 1$ and $\lim_{T\to 0}(-\log A(T))/T = 0$. (2) follows from $\lim_{T\to\infty} B(T)/T = 0$ and $\lim_{T\to\infty}(-\log A(T))/T = 2\alpha\bar{\kappa}/(\alpha+h)$ (standard CIR long-rate formula, Cox et al. [11]). (3) follows from the monotonicity of $B(T)$ in T and the fact that the yield is an affine function of κ_t. □ □

A.5 Economic Interpretation

Table 4 translates the three yield curve shapes into Islamic monetary policy language.

Table 4: κ-yield curve shapes and monetary interpretation.

Shape	Condition	Monetary signal
Normal (upward)	$\kappa_t < \bar{\kappa}$	Current convergence below trend; market expects tightening
Inverted (down)	$\kappa_t > \bar{\kappa}$	Convergence unusually fast; stress or anomaly
Flat	$\kappa_t = \bar{\kappa}$	Long-run equilibrium; neutral conditions

These shapes map directly onto the conventional yield curve taxonomy (normal/inverted/flat) with the Islamic interpretation: the κ-yield curve inverts not because markets expect *interest rate cuts*, but because markets expect *convergence to slow* as some structural uncertainty develops. This is a fundamentally different economic signal from the conventional inverted yield curve — it reflects real asset pricing uncertainty rather than monetary tightening expectations.

A.6 Calibration

Under the CIR-κ model, three parameters must be estimated: α, $\bar{\kappa}$, and σ_κ. Calibration uses the cross-section of implied $\hat{\kappa}$ values across instruments with different expected horizons $T^{(i)}$:

$$(\alpha^*, \bar{\kappa}^*, \sigma_\kappa^*) = \arg\min_{\alpha,\bar{\kappa},\sigma_\kappa} \sum_i \left[y_\kappa(T^{(i)};\kappa_t^{\text{obs}}) - y_\kappa^{\text{model}}(T^{(i)};\alpha,\bar{\kappa},\sigma_\kappa) \right]^2 \tag{25}$$

For the Baraka Protocol MVP with a single instrument ($T = 1/\hat{\kappa} \approx 12$ years), $\bar{\kappa} \approx 0.083$ is directly implied from the simulation empirical estimate (Section 7). The full stochastic calibration requires the multi-instrument deployment described in Section 5.

A.7 Relationship to CIR

Table 5 provides the formal correspondence between the conventional CIR interest rate model and the CIR-κ model.

Table 5: CIR vs. CIR-κ: parameter correspondence.

CIR element	CIR convention	CIR-κ analog
State variable	Short rate r_t	Convergence intensity κ_t
Long-run mean	$\bar{r}$	$\bar{\kappa}$
Mean-reversion speed	α	α
Volatility	σ_r	σ_κ
Positivity condition	$2\alpha\bar{r} \geq \sigma_r^2$	$2\alpha\bar{\kappa} \geq \sigma_\kappa^2$
Bond price	$A(\tau)e^{-B(\tau)r_t}$	$A(\tau)e^{-B(\tau)\kappa_t}$
Short rate	r_t (riba content)	κ_t (riba-free)
Economic meaning	Time preference	Convergence speed

The CIR-κ model is the CIR model with the riba content removed: the state variable κ_t (convergence intensity, an event-rate parameter) replaces r_t (short rate, a time-preference-compensation parameter). All the mathematical machinery of CIR — bond pricing, yield curve construction, risk-neutral calibration — carries over intact. The Islamic innovation is entirely in the economic interpretation of the state variable, not in the mathematics.

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